

E-commerce Analytics Case Study

Swiggy online ordering platform: sales, customers, products and trends

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Organisation	Davine Technologies - Data Analyst Internship
Activity	E-commerce Analytics Case Study (Data to Report)
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Tools	Excel / Python (Pandas), SQL, Power BI, GitHub
GitHub repository	https://github.com/aksharadeshmukh1311-crypto/swiggy-analytics
Power BI file	swiggy project.pbix (submitted with this report)



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1. Executive Summary and Portfolio Summary

This project analyses Swiggy restaurant, city and customer data covering about 149K records across 821 cities (2017-2020). The data was cleaned, analysed with SQL and Python, and presented in a 5-page Power BI dashboard (Overview, User Performance, City Overview, Restaurant Analysis, Insights). The aim is to show where sales come from, who the key customers are, and what Swiggy should do next.

At a glance

149K Orders count	100K Users	821 Cities	963.8M Total sales
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Headline findings

- **Top cities:** Tirupati leads all-time sales (43M), then Electronic City, Bangalore (29M), Baner, Pune (27M), Raipur (23M).
- **Customers:** Users aged 22-24 are the largest group (peak at about 2.1K users); students are 52.94% of the occupation mix on the 2020 female view.
- **Menu mix:** Veg leads on volume (156K vs 140K Non-Veg), but Non-Veg has the higher average price (231.81 vs 182.11).
- **Trend:** Quantity rose sharply from 2017 to a peak in 2018, then fell in 2019 and 2020.
- **Concentration:** The top 10% customers measure is 726.44M, about 75% of total sales of 963.83M.

Short Portfolio Summary

An end-to-end e-commerce analytics case study on about 149K Swiggy records across 821 cities (2017-2020). I cleaned the data with Python/Pandas, analysed it with SQL and Python, built a 5-page interactive Power BI dashboard, and turned the findings into business insights and recommendations. Sales are concentrated in a few cities, young students are the core customers, and a small top customer segment drives a large share of revenue.

Resume bullet: Built an end-to-end analytics project on 149K Swiggy records (821 cities, 2017-2020): cleaned data in Python/Pandas, analysed with SQL, designed a 5-page interactive Power BI dashboard,

and recommended a student-focused campaign and a VIP programme based on customer and city sales patterns.

2. Business Problem and Objectives

Business problem: As the data analyst for an online ordering (e-commerce) platform, I was asked by management to give a complete view of business performance. Swiggy's management needs to understand which cities, restaurants, cuisines and customer segments drive sales, and why growth slowed after 2018, so marketing and expansion budget can be focused on the highest-value areas.

Objectives

- Identify top cities and restaurants by sales, users and ratings.
- Profile customers by age, gender, marital status and occupation.
- Measure year-over-year change in sales and quantity (2017-2020).
- Compare Veg, Non-Veg and Others on volume and average price.
- Measure how concentrated revenue is in the top 10% of customers.
- Turn findings into clear, actionable recommendations.

3. Dataset and Data Collection

Source: Swiggy restaurant / order / user dataset used in the Power BI project (swiggy project.pbix). The raw files were collected into one folder, merged on the common keys (city, restaurant, user) and organised into the structure shown in Appendix C.

Fields used in the analysis (as seen in the dashboard)

Group	Fields
Location	City (821 unique values)
Restaurant	Name, Cuisine, Rating, Rating Count, Veg/Non-Veg/Others
Transaction	Price, Sales (Amount), Quantity, Year, Orders
User	Age, Gender, Marital Status (Single/Married), Occupation (Student, Employee, Self Employed, House wife)

Dashboard headline metrics

Metric	Value	Metric	Value
Orders count	149K (148,577)	Total cities	821
User count	100K	Total sales	963,829,618
Rating count	61K	Restaurant count	148K
Avg price - Veg / Non-Veg / Others	182.11 / 231.81 / 19.86	Current_YR_Sale / Previous_YR_Sale	141M / 332M

4. Analytics Workflow and Tools

Stage	What was done	Tool
1. Business problem	Defined questions and KPIs	Documentation
2. Data collection	Gathered and organised raw files	Excel / CSV
3. Cleaning	Fixed types, duplicates, missing values, outliers	Python (Pandas) / Excel
4. EDA and analysis	Aggregations, YoY growth, segmentation, Pareto check	SQL, Python
5. Visualisation	5-page interactive dashboard with DAX measures	Power BI
6. Insights and report	Findings, recommendations, PDF report	PDF, Markdown
7. Portfolio	Repository, README, resume bullet	GitHub

5. Data Cleaning

Cleaning was done with Python/Pandas (full script in Appendix A). Each step protects the accuracy of the dashboard numbers.

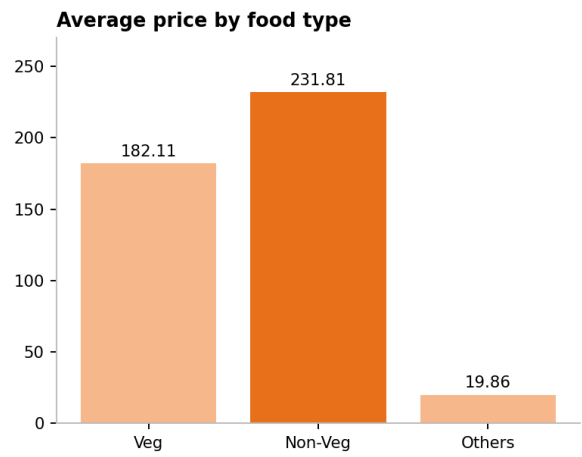
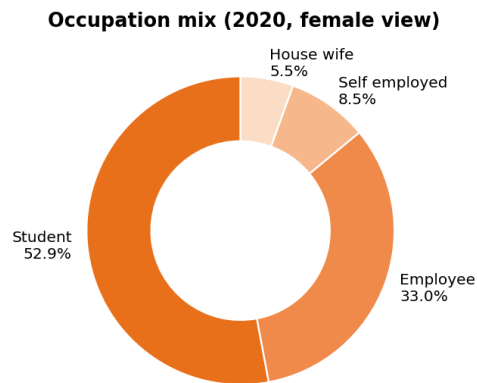
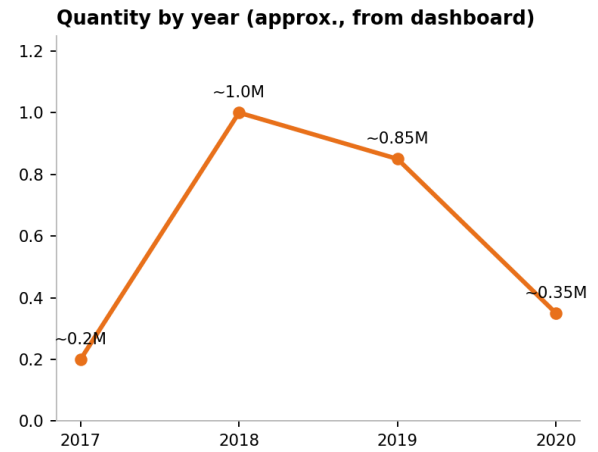
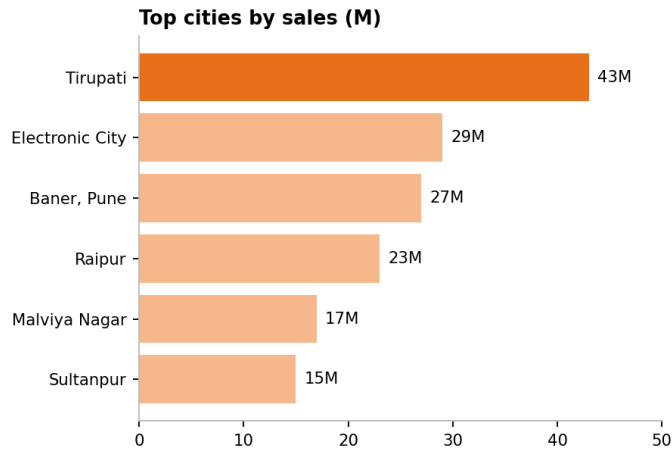
Step	Action	Why it matters
Standardise text	Trim spaces, consistent casing for city, cuisine, gender, occupation	Prevents one city or cuisine appearing as several values
Fix data types	Convert price, sales, quantity, rating, age to numeric; year to integer	Enables correct sums and averages
Duplicates	Drop exact duplicate rows	Avoids double-counting orders and sales
Missing values	Drop rows without city or sales; fill missing rating with median	Keeps KPIs reliable
Invalid records	Remove negative sales or price; keep age within 10-90	Removes impossible values
Outliers	IQR rule on sales; flagged, not deleted blindly	Extreme values may be genuine large orders
Export	Save data/cleaned/swiggy_cleaned.csv	Single clean source for SQL and Power BI

6. Exploratory Data Analysis and Analysis (SQL and Python)

The analysis answers each objective with a query or a Pandas aggregation. Full SQL is in Appendix B and Python in Appendix A.

Question	Method	Output used in
Top cities by sales	GROUP BY city, SUM(sales), ORDER BY DESC	City Overview, Insights
Year-over-year change	SUM(sales) by year with LAG() window function	Overview (Quantity by Year)
Veg vs Non-Veg	COUNT, SUM(sales), AVG(price) by category	Overview, Restaurant Analysis
Age, gender, occupation	Segment counts and sales by group	User Performance
Top 10% customers	NTILE(10) on customer sales, share of total	Overview, Insights
Top cuisines and restaurants	COUNT DISTINCT restaurants by cuisine	Restaurant Analysis

Analysis highlights



Source: Swiggy Power BI dashboard. Year and occupation values are read from the visuals.

7. Power BI Dashboard (5 pages)

The dashboard (file: swiggy project.pbix) has a consistent Swiggy-orange theme, a side navigation bar with icons, and slicers for Amount/Quantity, Top N, Year, Gender and City.

Page 1 - Overview. KPI cards (orders, users, rating count), average price for Veg, Non-Veg and Others, a Top N city bar chart (Electronic City leads with 0.36M quantity) and a Quantity by Year line chart.



Figure 1: Overview page

Page 2 - User Performance. Year slicer (2017-2020) and Gender toggle. Shows Marital Status (Single 41M, Married 16M), Occupation donut (Student 32M / 52.94%, Employee 20M / 33.04%, Self Employed 5M / 8.48%) and Users by Age (peak around age 23).

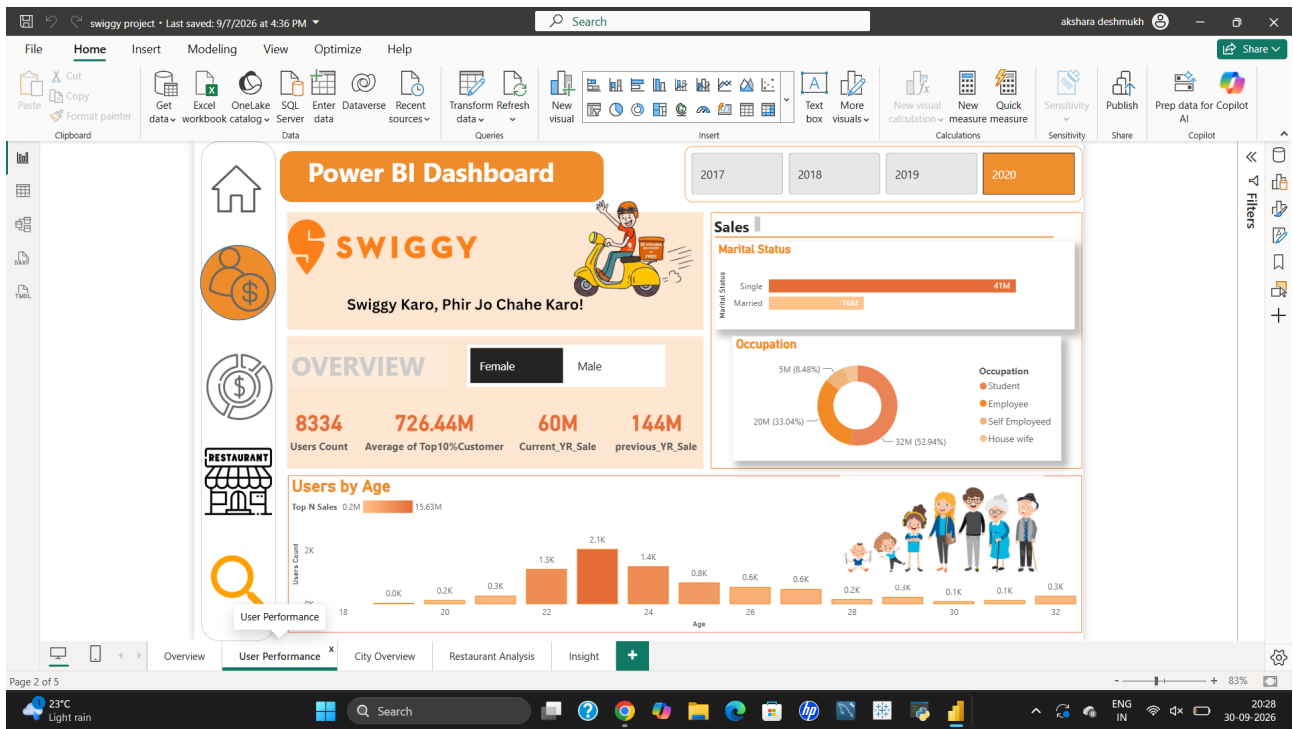


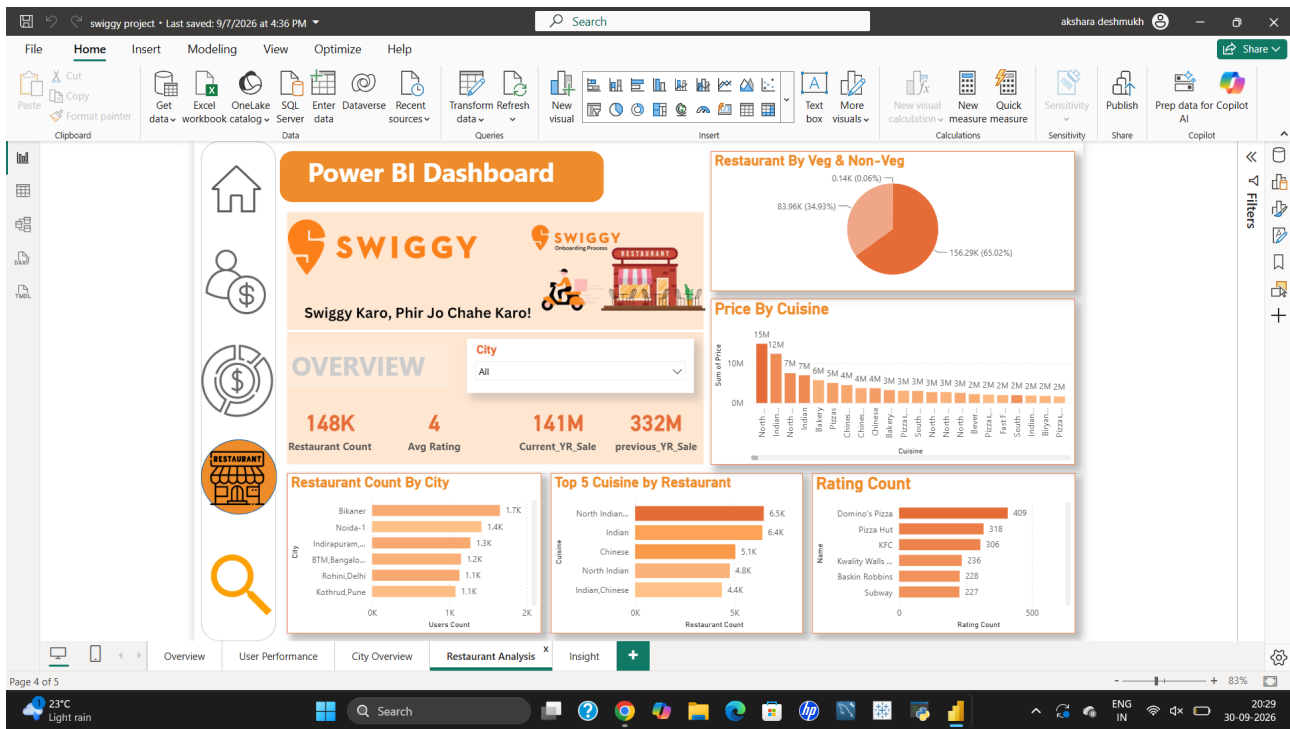
Figure 2: User Performance page (2020, Female filter)

Page 3 - City Overview. City slicer, a city table (sales, orders, current-year sales) and three bar charts for Sales, Users Count and Rating Count. Tirupati leads sales (43M); Bikaner leads users (1.7K) and ratings (850).

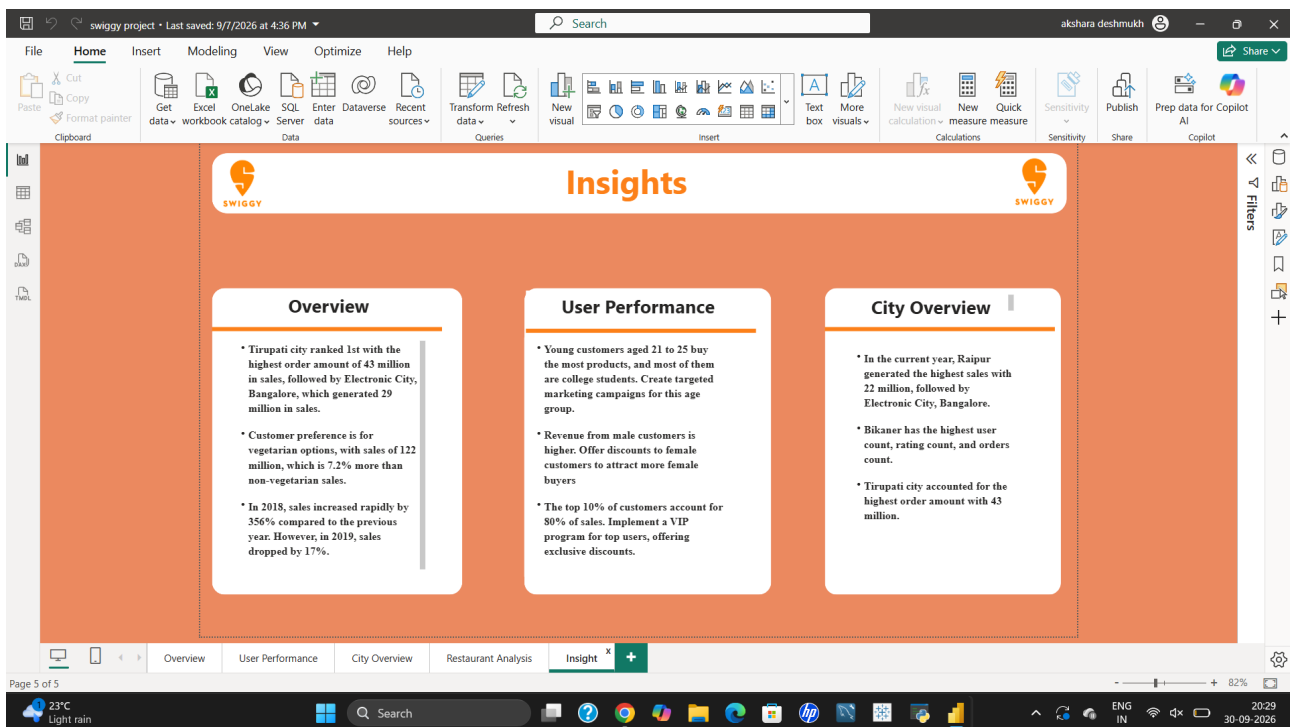


Figure 3: City Overview page

Page 4 - Restaurant Analysis. Veg / Non-Veg pie, Price by Cuisine, Restaurant Count by City, Top 5 Cuisines (North Indian 6.5K, Indian 6.4K, Chinese 5.1K) and top rated-count restaurants (Domino's 409, Pizza Hut 318, KFC 306).



Page 5 - Insights. Written findings and recommendations for Overview, User Performance and City Overview.



8. Key Business Insights (8 findings)

1. Sales are concentrated in a few cities Tirupati 43M, Electronic City 29M, Baner 27M and Raipur 23M, out of 963.8M total across 821 cities.	2. Bikaner leads engagement, not sales Highest users (1.7K) and rating count (850), yet it is not in the top sales list: a monetisation gap.
3. Young students drive demand Users peak at ages 22-24; students are 52.94% of the occupation mix (2020 female view).	4. Singles spend far more than married users Single 41M vs Married 16M (2020, female view).
5. Veg wins on volume, Non-Veg on price 156K vs 140K volume; average price 182.11 for Veg vs 231.81 for Non-Veg.	6. Growth peaked in 2018, then declined Quantity about 0.2M (2017) to 1.0M (2018), down to about 0.35M (2020). Current_YR 141M vs Previous_YR 332M.
7. Revenue depends on a small top segment The top 10% customer measure is 726.44M, about 75% of total sales.	8. Chain restaurants lead engagement Domino's, Pizza Hut and KFC have the highest rating counts.

9. Recommendations (in priority order)

Priority	Recommendation	Based on	KPI to track
High	Run campaigns for the 21-25 student segment (student offers, campus hubs, meal combos).	Insight 3	Orders from ages 18-25
High	Launch a VIP programme with exclusive discounts for the top 10% customers.	Insight 7	Repeat orders of top customers
High	Investigate the post-2018 decline (competition, discounts, data coverage) before scaling spend.	Insight 6	Sales vs previous year
Medium	Strengthen restaurant partnerships in Tirupati, Electronic City, Baner and Raipur.	Insight 1	Sales per city
Medium	Monetise Bikaner: test higher-value restaurants and promotions for its large user base.	Insight 2	Sales per user in Bikaner
Medium	Promote Non-Veg premium items and bundle them with Veg options.	Insight 5	Average order price
Medium	Offer targeted discounts to grow the female customer share (check male and female totals first).	User Performance page	Share of female users

10. Limitations and Next Steps

- The dataset covers 2017-2020 only; results may not reflect current market conditions.
- 2020 may be a partial year, which affects year-over-year comparisons.
- Customer segments (gender, occupation) are shown for a single filter view in the screenshots.
- Next: add a profit or margin field, run a cohort/retention analysis, and build a simple forecast of sales.

11. Project Documentation

The workflow followed was: Data, Cleaning, Analysis, Visualization, Insights, Recommendations, Report. Every stage is documented:

Deliverable	Where to find it
GitHub repository	Cover page link (README, report, code and screenshots)
Business problem and objectives	Section 2
Data cleaning and preparation	Section 5 and Appendix A (clean_and_eda.py)
SQL / Python analysis	Section 6, Appendix A and Appendix B (analysis.sql)
Interactive Power BI dashboard	Section 7 (5 page screenshots) and swiggy project.pbix
Key business insights	Section 8
Recommendations	Section 9
Project documentation	This section and Appendix C (README)
Portfolio summary	Section 1

How to reproduce

- Place the raw dataset next to clean_and_eda.py and set the file and column names at the top of the script.
- Run the script to create the cleaned CSV and print the EDA outputs.
- Load the cleaned CSV into a SQL database and run analysis.sql.
- Open swiggy project.pbix in Power BI Desktop and refresh the data source.

12. Recruiter's Review of this Project

I reviewed the project from a recruiter's point of view. A recruiter usually checks six things in a few minutes:

Recruiter question	How this project answers it
Is the business problem clear?	Section 2 states the problem, objectives and the decisions the analysis supports.
Can the analysis be reproduced?	Python and SQL code is included, with the steps to run it.
Is there visual proof?	Five dashboard page screenshots are shown in Section 7.
Are the insights specific?	Each insight in Section 8 is linked to numbers from the dashboard.
Does it lead to action?	Section 9 gives prioritised recommendations tied to the insights.
Is it honest about limits?	Section 10 lists data limitations (2017-2020 only, 2020 may be partial).

Planned improvements to make the portfolio stronger

- Rename the card 'Average of Top10%Customer' to 'Top 10% Customer Sales', since it is a sales total and not an average.
- Show the average rating with two decimals instead of '4'.
- Add a data dictionary and the cleaned dataset (or a sample) to the repository.
- Add a short Jupyter notebook that shows the EDA charts.
- Add a profit or margin field, a retention analysis and a simple sales forecast.

Appendix A: Python script (clean_and_eda.py)

```
"""Swiggy project - data cleaning and EDA.
Adjust the COLS mapping to match the column names in your raw file."""
import pandas as pd

RAW = "swiggy_raw.csv"
OUT = "swiggy_cleaned.csv"

# Left: name used below | Right: name in YOUR raw file (edit these)
COLS = {
    "city": "City", "restaurant": "Name", "cuisine": "Cuisine",
    "rating": "Rating", "rating_count": "Rating_Count", "price": "Price",
    "diet": "Veg_or_Non_Veg", "sales": "Sales", "quantity": "Quantity",
    "year": "Year", "age": "Age", "gender": "Gender",
    "marital": "Marital_Status", "occupation": "Occupation",
    "user_id": "User_ID",
}

df = pd.read_csv(RAW)
df = df.rename(columns={v: k for k, v in COLS.items() if v in df.columns})
print("Raw shape:", df.shape)

# 1. Standardise text columns
for c in ["city", "restaurant", "cuisine", "diet", "gender",
          "marital", "occupation"]:
    if c in df.columns:
        df[c] = df[c].astype(str).str.strip().str.title()

# 2. Fix data types
for c in ["rating", "rating_count", "price", "sales", "quantity", "age"]:
    if c in df.columns:
        df[c] = pd.to_numeric(df[c], errors="coerce")

# 3. Missing values and duplicates
print("Missing values:\n", df.isna().sum())
df = df.drop_duplicates()
df = df.dropna(subset=["city", "sales"])
if "rating" in df.columns:
    df["rating"] = df["rating"].fillna(df["rating"].median())

# 4. Remove invalid records
df = df[(df["sales"] >= 0) & (df["price"] >= 0)]
df = df[df["age"].between(10, 90) | df["age"].isna()]

# 5. Outlier check (IQR) on sales - flag, do not delete blindly
q1, q3 = df["sales"].quantile([0.25, 0.75])
iqr = q3 - q1
df["sales_outlier"] = ~df["sales"].between(q1 - 1.5 * iqr, q3 + 1.5 * iqr)
print("Sales outliers flagged:", int(df["sales_outlier"].sum()))

df.to_csv(OUT, index=False)
print("Cleaned shape:", df.shape)

# ----- EDA -----
print("\nTop 10 cities by sales")
print(df.groupby("city")["sales"].sum().nlargest(10))

print("\nQuantity by year")
qty = df.groupby("year")["quantity"].sum()
print(qty)
print("YoY % change:\n", (qty.pct_change() * 100).round(1))

print("\nAvg price and count by diet type")
print(df.groupby("diet").agg(count=("price", "size"),
                             avg_price=("price", "mean")).round(2))

print("\nUsers by age (top 10)")
print(df.groupby("age")["user_id"].nunique().nlargest(10))

print("\nSales by occupation and gender")
print(df.groupby("occupation")["sales"].sum().sort_values(ascending=False))
print(df.groupby("gender")["sales"].sum())

# Top 10% customers share of sales (Pareto check)
cust = df.groupby("user_id")["sales"].sum().sort_values(ascending=False)
top_n = max(1, int(len(cust) * 0.10))
share = cust.head(top_n).sum() / cust.sum() * 100
print(f"\nTop 10% customers contribute {share:.1f}% of total sales")
```

Appendix B: SQL queries (analysis.sql)

```
-- Swiggy project - analysis queries (MySQL / PostgreSQL compatible).
-- Table: swiggy_data(city, restaurant_name, cuisine, rating, rating_count,
-- price, veg_nonveg, sales, quantity, order_year, user_id, age, gender,
-- marital_status, occupation) -- rename to match your table

-- 1. Top 10 cities by sales
SELECT city, SUM(sales) AS total_sales, COUNT(*) AS orders
FROM swiggy_data GROUP BY city
ORDER BY total_sales DESC LIMIT 10;

-- 2. Year-over-year sales growth
WITH yearly AS (
    SELECT order_year, SUM(sales) AS total_sales
    FROM swiggy_data GROUP BY order_year)
SELECT order_year, total_sales,
    ROUND(100.0 * (total_sales - LAG(total_sales) OVER (ORDER BY order_year))
    / LAG(total_sales) OVER (ORDER BY order_year), 1) AS yoy_pct
FROM yearly ORDER BY order_year;

-- 3. Veg vs Non-Veg: volume, sales, average price
SELECT veg_nonveg, COUNT(*) AS records, SUM(sales) AS total_sales,
    ROUND(AVG(price), 2) AS avg_price
FROM swiggy_data GROUP BY veg_nonveg;

-- 4. Users by age group
SELECT CASE WHEN age < 18 THEN 'Under 18'
    WHEN age BETWEEN 18 AND 20 THEN '18-20'
    WHEN age BETWEEN 21 AND 25 THEN '21-25'
    WHEN age BETWEEN 26 AND 30 THEN '26-30'
    ELSE '31+' END AS age_group,
    COUNT(DISTINCT user_id) AS users, SUM(sales) AS total_sales
FROM swiggy_data GROUP BY 1 ORDER BY users DESC;

-- 5. Sales by gender and occupation
SELECT gender, occupation, SUM(sales) AS total_sales
FROM swiggy_data GROUP BY gender, occupation
ORDER BY total_sales DESC;

-- 6. Top 10% customers: share of total sales
WITH cust AS (
    SELECT user_id, SUM(sales) AS s,
        NTILE(10) OVER (ORDER BY SUM(sales) DESC) AS decile
    FROM swiggy_data GROUP BY user_id)
SELECT ROUND(100.0 * SUM(CASE WHEN decile = 1 THEN s END) / SUM(s), 1)
    AS top10pct_share
FROM cust;

-- 7. Top 5 cuisines by restaurant count
SELECT cuisine, COUNT(DISTINCT restaurant_name) AS restaurants
FROM swiggy_data GROUP BY cuisine ORDER BY restaurants DESC LIMIT 5;

-- 8. Cities with the highest user and rating counts
SELECT city, COUNT(DISTINCT user_id) AS users, SUM(rating_count) AS ratings
FROM swiggy_data GROUP BY city ORDER BY users DESC LIMIT 10;
```

Appendix C: GitHub README (README.md)

Swiggy Sales & Customer Analytics - End-to-End Data Analytics Project

An end-to-end analytics project on Swiggy restaurant, city and customer data: data collection, cleaning, SQL/Python analysis, a 5-page Power BI dashboard, business insights and recommendations.

1. Business Problem

Swiggy's business team needs to know where sales come from, who the most valuable customers are, and which cities and cuisines to prioritise.

The project answers:

1. Which cities and restaurants drive the most sales, users and ratings?
2. Who are the core customers (age, gender, occupation, marital status)?
3. How have sales and quantity changed year over year (2017-2020)?
4. How do Veg, Non-Veg and Other categories compare on volume and price?
5. How concentrated is revenue in the top 10% of customers?

2. Dataset

- Source: Swiggy restaurant / order / user dataset used in the Power BI project
- Coverage: about 149K records, 821 cities, years 2017-2020
- Key fields: City, Restaurant Name, Cuisine, Rating, Rating Count, Price, Veg/Non-Veg, Sales (Amount), Quantity, Year, User Age, Gender, Marital Status, Occupation

3. Tools Used

Stage	Tool
Cleaning	Python (Pandas) / Excel
Analysis	SQL, Python
Visualisation	Power BI (DAX measures)
Documentation	Markdown, PDF, GitHub

4. Workflow

Raw data -> Cleaning -> EDA -> SQL/Python analysis -> Power BI -> Insights
-> Report -> GitHub.

5. Dashboard Pages

1. Overview - orders, users, avg price by Veg/Non-Veg/Others, top 10 cities, quantity by year
2. User Performance - age, gender, marital status, occupation, year slicer
3. City Overview - sales, users and ratings by city, city slicer
4. Restaurant Analysis - veg vs non-veg mix, cuisines, top restaurants
5. Insights - written findings and recommendations

6. Key Insights

- Tirupati leads sales (43M), followed by Electronic City, Bangalore (29M).
- Customers aged 21-25 (mostly students) are the largest user group.
- Veg options lead on sales; Non-Veg has the higher average price.
- Sales grew sharply in 2018 and declined in 2019 and 2020.
- Revenue is heavily concentrated in the top 10% of customers.

7. Recommendations

1. Run targeted campaigns for the 21-25 student segment.
2. Launch a VIP programme for top-spending customers.
3. Offer discounts to attract more female customers.
4. Expand partnerships in high-sales cities (Tirupati, Raipur, Bangalore).
5. Investigate the post-2018 decline before scaling marketing spend.

8. Repository Contents

```
swiggy-analytics/  
|-- README.md                project overview  
|-- Swiggy_Analytics_Project_Report.pdf full business report  
|-- clean_and_eda.py         data cleaning and EDA (Python)  
|-- analysis.sql             analysis queries (SQL)  
|-- screenshots (PNG)        5 Power BI dashboard pages  
...
```

Power BI file: swiggy project.pbix (submitted with the report; not stored here because of its file size).

9. How to Run

1. Put the raw dataset in the same folder as clean_and_eda.py and set the file name and column names at the top of the script.
2. Run: python clean_and_eda.py (creates the cleaned CSV)
3. Load the cleaned CSV into a SQL database and run analysis.sql.
4. Open swiggy project.pbix in Power BI Desktop and refresh the data source.

10. Author

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GitHub: <https://github.com/aksharadeshmukh1311-crypto/swiggy-analytics>